The due date for submitting this assignment has passed.
Due on 2026-03-25, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-03-25, 07:24 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

Multiple Select Questions (MSQ):

1) A function $T : V \rightarrow W$ between two vector spaces V and W is said to be a linear transformation if the following conditions hold: ***1 point***

Condition 1: $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$.
Condition 2: $T(cv) = cT(v)$ for all $v \in V$ and $c \in \mathrm{R}$.

Consider the following function:

$$T : \mathrm{R}^2 \rightarrow \mathrm{R}$$
$$T(x, y) = \{ \begin{matrix} 3x & \text{if } y = 0 \\ 4y & \text{if } y = 0 \end{matrix}$$

Which of the following statements is true?

- [ ] Condition 1 holds.
- [x] Condition 1 does not hold.
- [x] Condition 2 holds.
- [ ] Condition 2 does not hold.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Condition 1 does not hold.
Condition 2 holds.

2) Suppose the matrix representation of a linear transformation $T : \mathrm{R}^3 \rightarrow \mathrm{R}^3$ with respect to ordered bases $\beta = \{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$ for the domain and $\gamma = \{(1, 0, 0), (0, 1, 0), (1, 0, 1)\}$ for the range, is $I_{3\times3}$, i.e., the identity matrix of order 3. Let A denote the matrix representation of the linear transformation T with respect to the standard ordered basis of R^3 for both domain and range. Which of the following are true? ***1 point***

- [ ] $A = I_{3\times3}$ i.e., identity matrix of order 3.
- [ ] A is a singular matrix
- [ ]

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

- [x] $det(A) = 1.$
- [ ] $det(A) = -1.$
- [x] $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

Yes, the answer is correct.

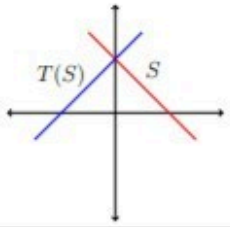
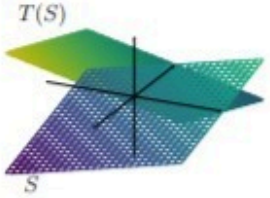
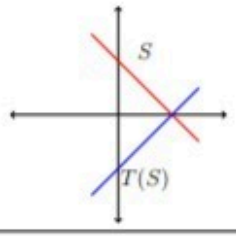
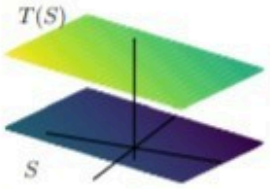
Score: 1

Accepted Answers:

$det(A) = 1.$

$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

3) Match the linear transformations and sets of vectors in column A with the images of those sets under the linear transformation in column B and the geometric representations of both sets in column C. ***1 point***

	Matrix form of linear transformation (Column A)		Image of the given set (Column B)		Geometric representations (Column C)
i)	$T:\mathbb{R}^3 \to \mathbb{R}^3$ $T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 2 & 3 \end{bmatrix}$ Set: $S = \{(x,y,z) \mid x+y+z=1\}$	a)	$T(S) = \{(x,y) \mid x-y=1\}$	1)	
ii)	$T:\mathbb{R}^3 \to \mathbb{R}^3$ $T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ Set: $S = \{(x,y,z) \mid x+y+z=1\}$	b)	$T(S) = \{(x,y,z) \mid x+y+z=3\}$	2)	
iii)	$T:\mathbb{R}^2 \to \mathbb{R}^2$ $T = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ Set: $S = \{(x,y) \mid x+y=1\}$	c)	$T(S) = \{(x,y) \mid x-y=-1\}$	3)	
iv)	$T:\mathbb{R}^2 \to \mathbb{R}^2$ $T = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ Set: $S = \{(x,y) \mid x+y=1\}$	d)	$T(S) = \{(x,y,z) \mid x+y-z=1\}$	4)	

Choose the correct option from the following.

- [ ] i $\to$ d $\to$ 2, ii $\to$ b $\to$ 4.

i → b → 4, ii → d → 2.

- [x] iii → a → 1, iv → c → 3
- [ ] iii → a → 3, iv → c → 1

No, the answer is incorrect.

Score: 0

Accepted Answers:

i → b → 4, ii → d → 2.
iii → a → 3, iv → c → 1

4) Consider a linear transformation $S : M_2(\mathrm{R}) \to M_2(\mathrm{R})$ such that $S(A) = A^T$. Let B be the matrix representation of S with respect to the ordered bases: ***1 point***

$$\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\}$$

of $M_2(\mathrm{R})$. Choose the set of correct options:

- [ ] The order of the matrix B is 2×2.
- [x] The order of the matrix B is 4×4.
- [x] The dimension of the row space of the the matrix B is 4.
- [ ] The dimension of the column space of the matrix B is 3.
- [ ] The nullity of the the matrix B is 1.
- [x] The rank of the the matrix B is 4.
- [x] S is surjective.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The order of the matrix B is 4×4.
The dimension of the row space of the the matrix B is 4.
The rank of the the matrix B is 4.

S is surjective.

5) Let $T : \mathrm{R}^n \to \mathrm{R}^m$ be a linear transformation and tet $\{v_1, v_2, \ldots, v_n\}$ be a basis of R^n. Suppose $\{T(v_1), T(v_2) \ldots, T(v_n)\}$ is a spanning set for R^m. Choose all the correct statements. ***1 point***

- [x] T is a surjective linear transformation.
- [ ] T is a linear isomorphism.
- [x] $m \leq n$.
- [x] For any basis $\{w_1, w_2, \ldots, w_n\}$ of R^n, the set $\{T(w_1), T(w_2), \ldots, T(w_n)\}$ is a spanning set for R^m.
- [ ] The matrix of the linear transformation T has rank equal to n.

Yes, the answer is correct.

Score: 1

Accepted Answers:

T is a surjective linear transformation.
$m \leq n$.
For any basis $\{w_1, w_2, \ldots, w_n\}$ of R^n, the set $\{T(w_1), T(w_2), \ldots, T(w_n)\}$ is a spanning set for R^m.

Comprehension Type Question:

Suppose a bread-making machine B makes 6 breads from 2 eggs, 3 (in hundreds) grams of wheat, and 1 (in hundred) grams of sugar. B also makes 8 breads from 3 eggs, 4 (in hundreds) grams of wheat, and 2 (in hundreds) grams of sugar, and 10 breads from 5 eggs, 5 (in hundreds) grams of wheat, and 3 (in hundreds) grams of sugar. Suppose the production of breads is a linear function of the amount of eggs, wheat (in hundreds), and sugar (in hundreds) used as raw ingredients. Based on the above data answer the following questions. Suppose x eggs, y (in hundreds) grams of wheat, and z (in hundreds) grams of sugar are used as the raw materials to produce $ax + by + cz$ number of breads. We can express this as follows:

$$T : \mathrm{R}^3 \rightarrow \mathrm{R}$$

$$T(x, y, z) = ax + by + cz$$

where the co-ordinates in R^3 denote the number of eggs, amount (in grams) of wheat (in hundreds), and amount (in grams) of sugar (in hundreds). Observe that T is a linear transformation.

Use the above information for Question 6, 7 and 8.

6) Choose the correct set of options from the the following. (MSQ) ***1 point***

- [ ] $Nullity(T) = 1$
- [x] $Rank(T) = 1$
- [x] $Nullity(T) = 2$
- [ ] $Rank(T) = 2$
- [ ] $Nullity(T) = 3$
- [ ] $Rank(T) = 3$
- [ ] T is neither one to one nor onto.
- [ ] T is one to one but not onto.
- [x] T is onto but not one to one.
- [ ] T is an isomorphism.

Yes, the answer is correct.

Score: 1

Accepted Answers:

$Rank(T) = 1$
$Nullity(T) = 2$
T is onto but not one to one.

7) Choose the set of correct statements. (MSQ) ***1 point***

- [ ] If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then 9 breads are produced.
- [x] If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then no bread is produced.
- [x] If only 3 (in hundreds) grams of wheat is used, then 6 breads are produced.

- [ ] If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then no bread is produced.
- [x] If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then 6 breads are produced.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then no bread is produced.
If only 3 (in hundreds) grams of wheat is used, then 6 breads are produced.
If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then 6 breads are produced.

8) How many breads are produced by the machine from 6 eggs, 10 (in hundreds) grams of wheat, and 5 (in hundreds) grams of sugar? (NAT)

20

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 20

1 point

9) Let T be a linear transformation from R^{10} to R^{12}. Suppose a basis for the null space of T has 3 vectors. How many linearly independent vectors are needed to form a basis for the range of T?

7

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 7

1 point

10) Let M be the set of all skew-symmetric matrices of order 7. Then M forms a vector space under matrix addition and scalar multiplication. Let T be a linear transformation from M to R defined by $T(A) = 6\ tr(A)$, where $tr(A)$ =trace of A. What is the nullity of T?

21

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 21.0

1 point

11) Let $T : \mathrm{R}^3 \longrightarrow \mathrm{R}^2$ be defined by $T(x, y, z) = (10x + 9y, 2z)$. Choose the correct options about T. ***1 point***

- [ ] The matrix of T is $\begin{bmatrix} 10 & 0 \\ 9 & 0 \\ 0 & 2 \end{bmatrix}$.
- [x] The matrix of T is $\begin{bmatrix} 10 & 9 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.
- [x] A basis for the null space of T is $\{(1, -\frac{10}{9}, 0)\}$.
- [ ] A basis for the null space of T is $\{(1, -\frac{10}{9}, 0), (0, 0, 2)\}$.
- [ ] A basis for the range of T is $\{(10, 10)\}$.
- [x] A basis for the range of T is $\{(10, 9), (0, 2)\}$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The matrix of T is $\begin{bmatrix} 10 & 9 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.
A basis for the null space of T is $\{(1, -\frac{10}{9}, 0)\}$.
A basis for the range of T is $\{(10, 9), (0, 2)\}$.